

## HUMAN ORGAN SYSTEM

In the previous classes we have studied human digestive and respiratory systems. Do you know how these systems are controlled? Which organ system is responsible for recognizing and remembering things? How do you understand things? How do waste products are secreted from the body? The body system that controls and coordinates for the function of all other organ systems is called Nervous System. For example, when nervous system gives signal to the excretory system it helps to remove metabolic wastes from the body. Let us explore the structure and functions of these amazing systems.

**In this Chapter you will learn about:**

- Nervous System (Central and Peripheral)
- Reflex Action
- Excretory System (Structure of Kidney and its Role in Excretion)

**All the students will be able to:**

- ✓ Describe the structure and functions of the nervous system.
- ✓ Describe the working of the nervous system through a model.
- ✓ Explain reflex action with an example.
- ✓ Differentiate between voluntary and involuntary actions they have experienced.
- ✓ Define excretion.
- ✓ Draw and label human excretory system.
- ✓ Describe the role of kidney in the excretion of waste.
- ✓ Investigate the possible causes of the malfunctioning of kidneys.
- ✓ Suggest techniques to cure problems of kidneys.

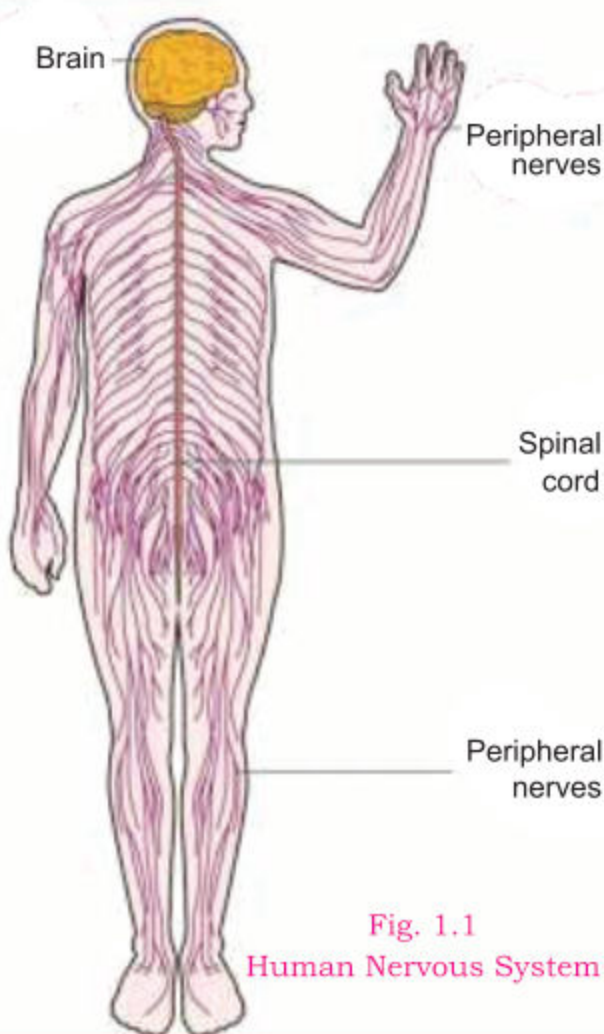


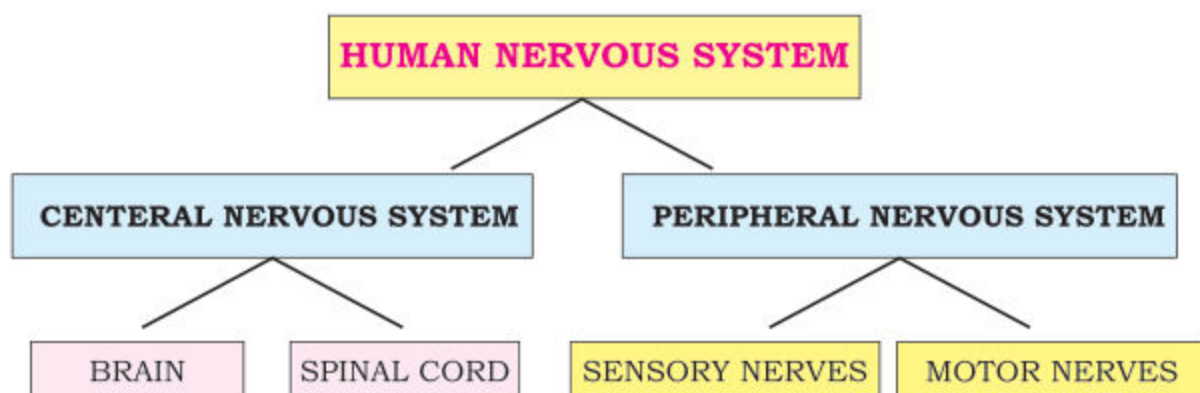
Fig. 1.1  
Human Nervous System

## NERVOUS SYSTEM (CENTRAL AND PERIPHERAL)

- Describe the structure and functions of the nervous system.

The nervous system is a network of nerves that links brain to every part of the body and allows them to work together. It is our body's communication system. It is divided in two parts:

1. The Central Nervous System (CNS)
2. The Peripheral Nervous System (PNS)



### 1. Central Nervous System (CNS)

The central nervous system is composed of the brain, spinal cord and nerves. It controls all the vital functions necessary for living such as breathing, maintaining body temperature, thinking, feelings and heartbeat.

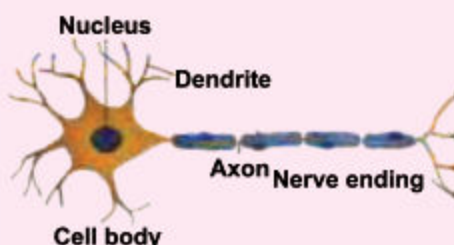
#### A) Brain

Have you ever seen a brain? Do you know that the structure of brain is very similar to the structure of walnut as shown in Fig. 1.2. Let's discuss the human brain.

Brain is enclosed in a hard-bony protective structure called cranium (skull). Brain is made up of soft nerve

#### DO YOU KNOW?

Nervous system has the responsibility to communicate, and coordinate for the body functions. It has specialized nerve cells called neurons, which are the basic structural and functional units or building blocks of the nervous system.



tissues and appears like a large pinkish grey walnut. Brain has folds and depressions which give it wrinkled appearance. It is the most amazing part of our body which serves as control centre of the human body. It receives messages from all parts of the body and gives orders to control the activities of different parts of the body. Brain is always working even while sleeping. Brain consists of three parts.

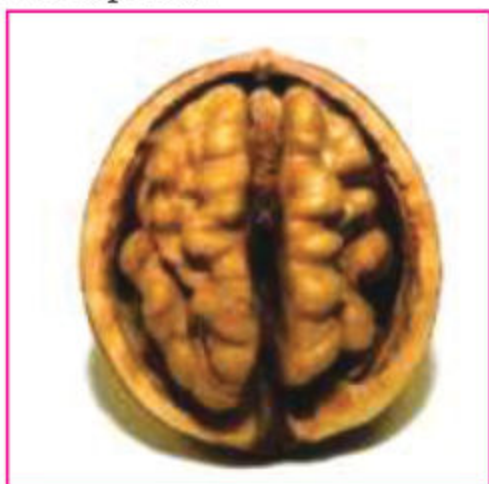


Fig.1.2: Walnut

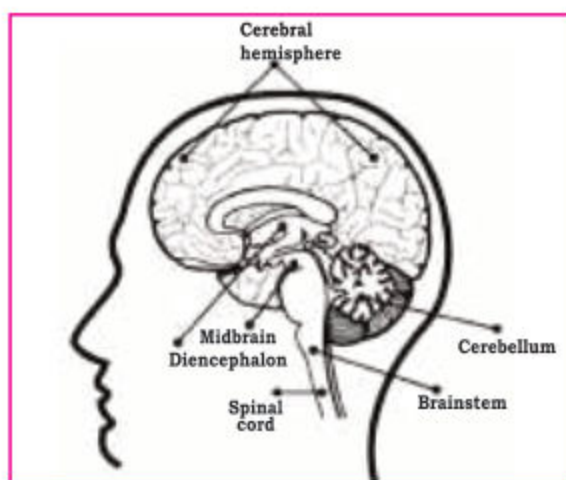


Fig.1.3:Parts of human brain

**i) Fore Brain or Cerebrum:** It is the largest part of the brain and comprises two hemispheres. Right hemisphere controls the left side of the body and left hemisphere controls the right side of the body. Cerebrum controls many actions like thinking, feelings, memory, hearing, seeing, speech, decision making.

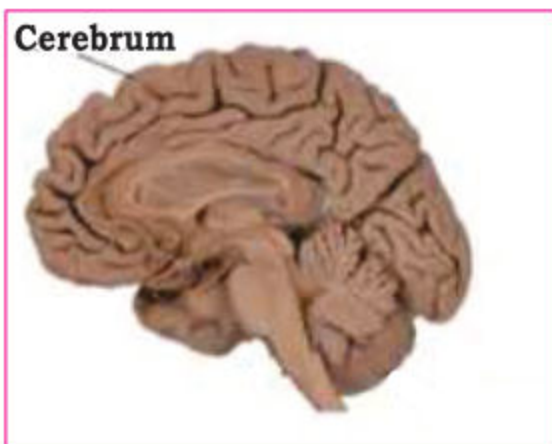


Fig.1.4: Fore Brain

**Thalamus:** It lies inside cerebrum and controls sensory functions.

**Hypo-thalamus:** It lies at the base of the thalamus and controls the body temperature, feeling of hunger and thirst.

**ii) Mid Brain:** The midbrain consists of the optic lobes. The optic lobes are four small bodies. These are concerned with visual reflexes, e.g. movement of the eyeballs.

**iii) Hind Brain:** The hind brain consists of three parts:

- **Pons**
- **Cerebellum**
- **Medulla oblongata**

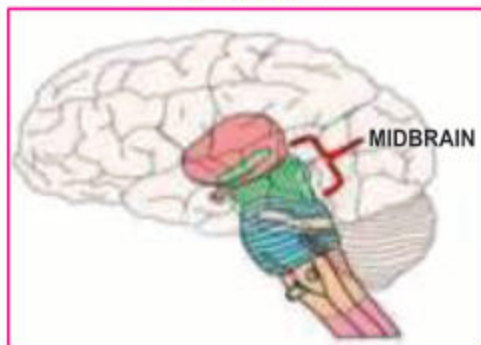
The **Pons** is a bridge-like structure which links different parts of the brain. It conveys signals from the medulla to the higher parts of the brain. Pons also controls facial expressions, sleeping and breathing.

The cerebellum lies dorsally behind the optic lobes. It is large, and its surface contains many folds. The cerebellum plays an important part in controlling muscular co-ordination, and especially in maintaining the body balance. The medulla Oblongata lies below the cerebellum. Its lower end narrows down gradually into the spinal cord.

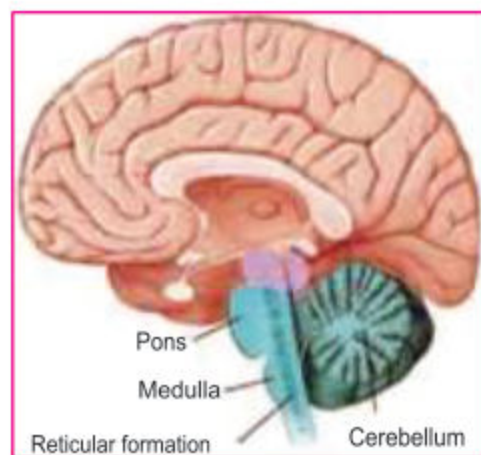
The medulla oblongata controls involuntary reflex actions such as blood pressure, heart beat, peristalsis, the rate of respiratory movements and the contraction and dilation of blood vessel.

## **B) Spinal Cord**

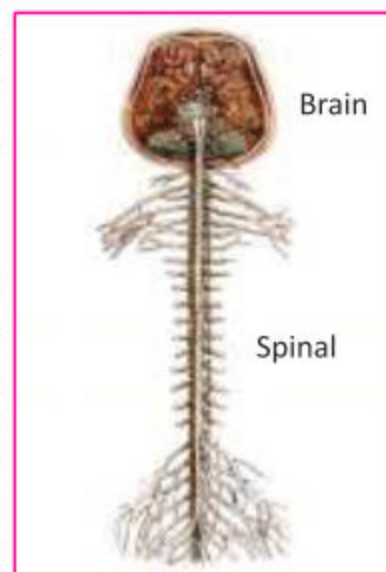
The spinal cord is a long piece of nerve tissue that runs from the brain through the backbone within the vertebral column. In fact it is the extension of the brain through the vertebral column. It receives sensory information from back parts of the body. The brainstem (Medulla Oblongata) connects the brain to the spinal cord while spinal cord



**Fig.1.5: Mid Brain**



**Fig.1.6: Hind Brain**



**Fig.1.7: Spinal Cord**

connects the brain to the body's nerves. Spinal cord helps regulate and control functions which are not under the control of persons' will such as breathing, heartbeat, blinking, and blood pressure.

### **C) Peripheral Nervous System**

The peripheral nervous system consists of nerves that branch out from the central nervous system to the rest of the body. It acts as the lines of communication between the Central Nervous System and the rest of the body.

Nerves are of two types, one type of nerves carries messages to the brain and spinal cord; these are called sensory nerves. The second type of nerves carries messages away from brain or spinal cord to the organs; these are known as motor nerves.

The function of the cerebral cortex can be understood by dividing it into four zones such as:

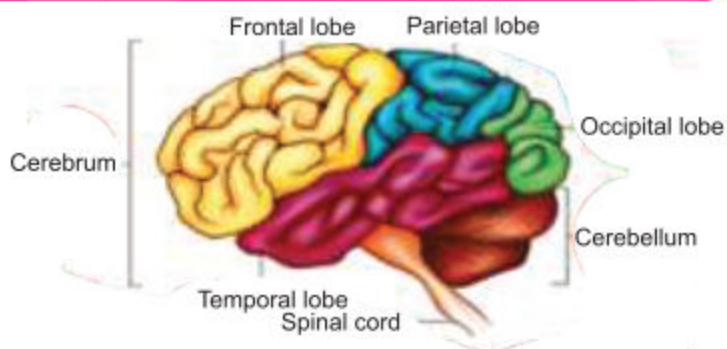
1. **The frontal lobe:** is responsible for initiating and coordinating motor movements; higher thinking skills, such as problem solving, planning, reasoning, movement, and organizing; and for many aspects of personality.
2. **The parietal lobe:** is involved with sensory processes, attention, and language. The right side of the parietal lobe helps in navigating spaces. The left side helps to understand spoken and written language including recognition of shapes and colors.
3. **The temporal lobe:** helps to process auditory information and integrate information from the other senses. Temporal lobe plays a role in short-term memory formation.
4. **Occipital lobe:** helps process visual information, including recognition of colours. All of these structures make up the fore brain.

In order to enhance your understanding about structure of nervous system associated with senses such as touch, pressure, temperature and pain, now perform this simple activity which will convert your abstract ideas into concrete understanding.

## ACTIVITY 1.1

### Understanding the Structure of Brain:

Studying a real human brain is not possible at this level. Teacher can help students to relate the learning with their own body. This activity may help students to visualize the major section of human brain. Teachers can provide a white tight cap and ask students to locate different zone of the brain on the cap.

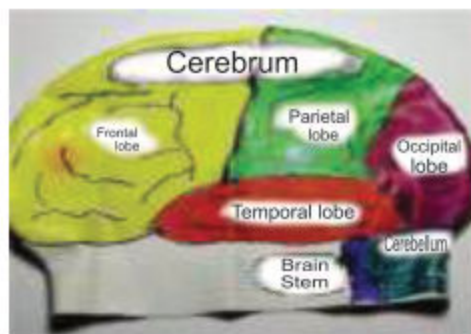


### What I need:

- A diagram of Brain as shown above (one for each group).
- A black marker and markers of different colours for drawing 4 zones, spinal cord and cerebellum (one set for each group)
- White cap made up of cloth/ swimming cap/ bath cap (1 for each group) (teacher can also prepare a paper/ cloth cap).
- Glue/ Scotch Tape.

### What to do:

1. With the help of a black marker, mark different zones on the cap as shown in the figure.
2. Use different colour markers to show different zones (frontal, parietal, lateral, occipital) cerebellum and medulla oblongata.
3. Teacher can wear the cap to demonstrate her/his brain portions.
4. One student from the group should wear the group's cap to show and explain other group members about different parts of brain. Turn by turn each group member will do the same.



### DO YOU KNOW?

The brain is responsible to process the information collected by the body and senses. It controls and regulates important processes, such as breathing. You should care for your brain and senses by wearing seatbelts and helmets while travelling.

## REFLEX ACTION

- ✓ Explain Reflex Action with an example.

The quick action in which brain is not involved is called a reflex action. If you accidentally touch a hot pot on your stove, you will immediately pull your hand away from the pot. This quick response is called a reflex action. Such types of actions are automatic and superfast performed by the body in response to a sensation. They

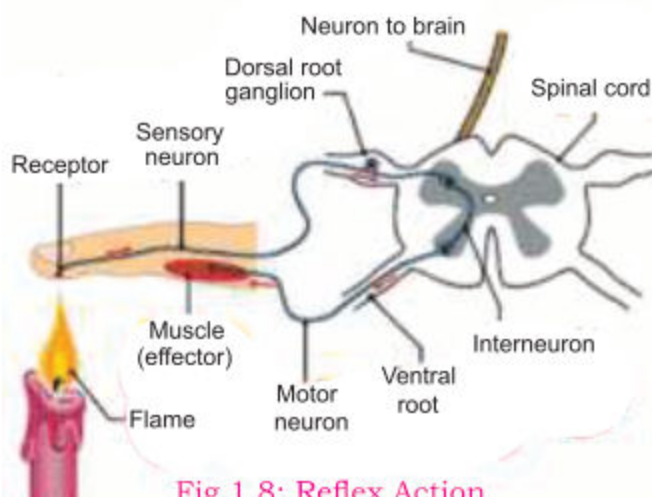


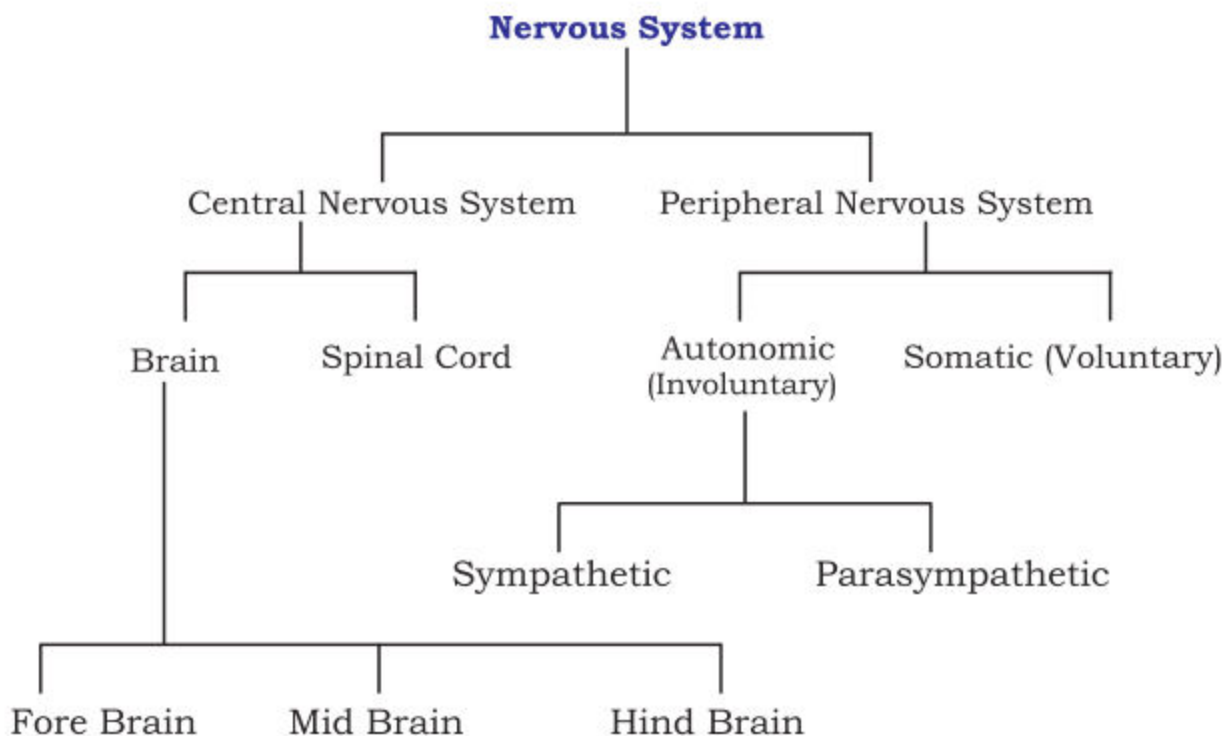
Fig.1.8: Reflex Action

are controlled by nerve messages that work even before person thinks about them. It means our all actions are not ordered by brain. Nerves in our hand detect the heat and shoot a message to nerve centre in the spinal cord. The nerve centre immediately sends back a message to the hand to draw itself away, this happens within fraction of a second. Closing the eyes quickly, if an object suddenly approaches it, withdrawing the hand when pricked, coughing and sneezing are natural reflex actions.

## DIFFERENCE BETWEEN VOLUNTARY AND INVOLUNTARY ACTIONS

- ✓ Differentiate between Voluntary and Involuntary Actions they have experienced.

| Voluntary or Conscious Action                                                                                                                                                                                                                                                                             | Involuntary or not under conscious control                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• These actions are performed according to our wish.</li><li>• We have experienced eating, bathing, walking, running and doing all that day to day activities that involve muscles with brain.</li><li>• Fore brain is responsible for voluntary actions.</li></ul> | <ul style="list-style-type: none"><li>• Most of the biological processes in animals are involuntary which occur without conscious control or will.</li><li>• Breathing, blinking of eyes, contraction of the heart, blood flow and digestion are involuntary actions because we cannot alter their course of action.</li><li>• Hind brain is responsible for involuntary actions.</li></ul> |



## EXCRETORY SYSTEM: STRUCTURE OF KIDNEY AND ITS ROLE IN EXCRETION

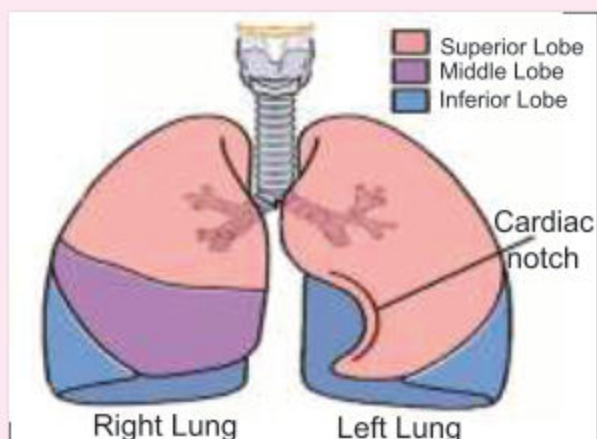
### ➤ Define Excretion

In the previous class you have studied human digestive and respiratory system. It is now time to recall your previous knowledge. Have you ever thought what happens to the undigested food and the carbon dioxide gas produced as part of digestion and respiration respectively? What will happen to our body if these waste materials remain in our body? How do we get rid of these waste materials? What happens to water that we drink? How do we eliminate liquid waste from our body? What is the function of kidneys? Let us explore.

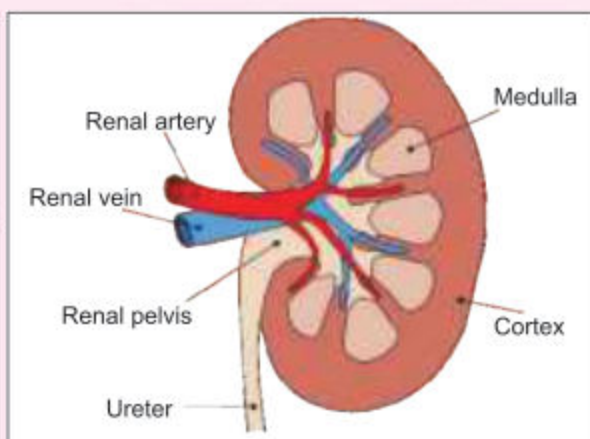
The excretory system is critical to survival. The trillions of living cells in the human body produce metabolic wastes which must be eliminated from the body. If the wastes aren't eliminated, it might lead to death. The process of eliminating these waste materials from the body is known as excretion of waste.

### DO YOU KNOW?

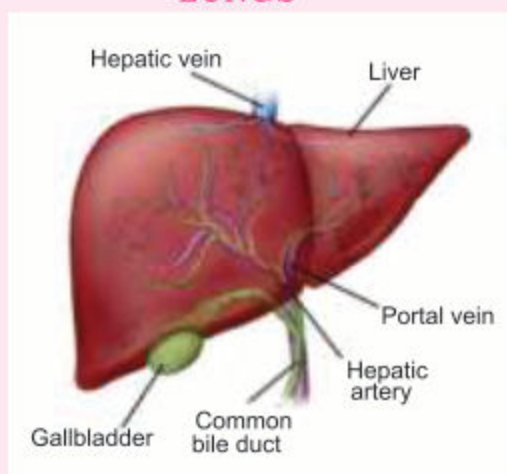
There are four organs which play an important role in eliminating waste material from our body. These are: 1. Lungs 2. Kidneys 3. Liver 4. Skin



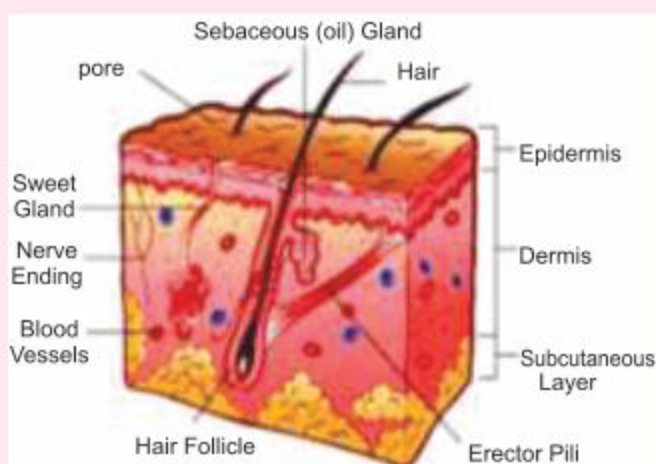
LUNGS



KIDNEY



LIVER



SKIN

### DO YOU KNOW?

What are the waste materials that being excreted from the human body?

- **Water**, is a product of cellular respiration and excess water absorbed from food in the large intestine.
- **Carbon dioxide**, which is a waste material from the process of cellular respiration.
- **Mineral salts** such as excess sodium chloride from food.
- **Urea** that is nitrogenous material produced from the breakdown of excess protein in the liver.

## THE HUMAN EXCRETORY SYSTEM

➤ Draw and label Human Excretory System

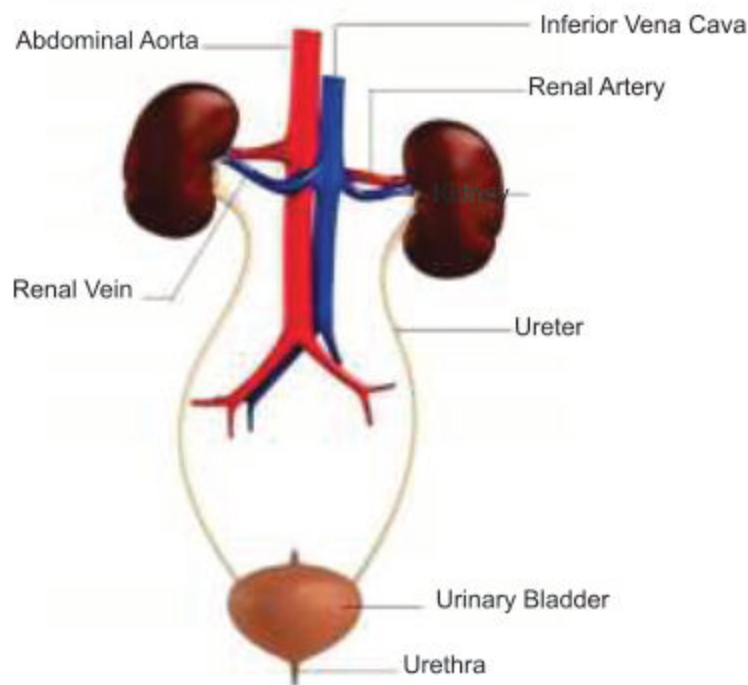


Fig. 1.9 The Human Excretory System

## THE ROLE OF KIDNEY

➤ Describe the role of kidney in the excretion of waste.

In the previous chapters you studied that as part of human digestive system, undigested food was expelled out from the body through large intestine. Similarly, during respiration, Carbon dioxide and water are eliminated through blood circulatory system, body transport system and respiratory system. It is important to note that all the systems in human body are connected and interrelated to perform body functions, so the different waste material especially excess water, metabolic wastes and salts are removed from body. Here, we will study in detail the main organ of excretory system that is kidney.

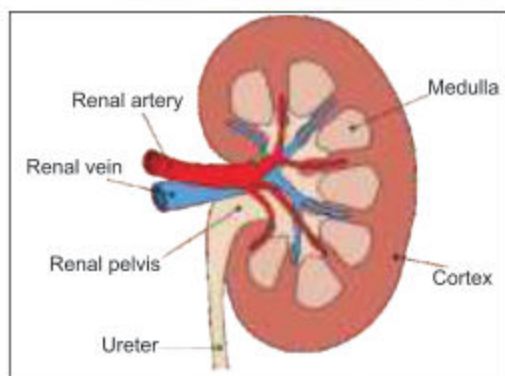


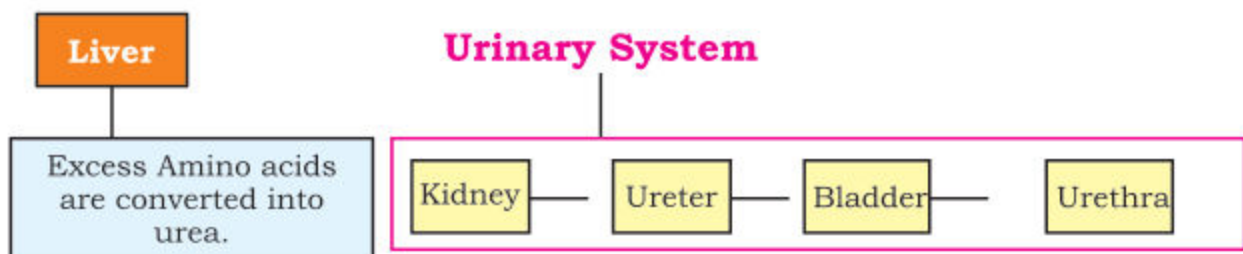
Fig.1.10: Kidneys

Kidneys are essential organs in human excretory system. There are two kidneys on the lower abdominal region, one on either side of the vertebral column, which are left kidney and right kidney. The kidneys are bean-shape and are reddish brown. When it is full, the excretory product eliminates out of the body through the urethra. This excretory product is discharged in the form of yellow-coloured liquid known as urine.

### How does Kidney Function:

The renal artery transports blood to the kidneys. This blood is rich in water, urea and dissolved mineral salts. Each kidney has about a million tiny units called nephrons. Nephrons take in blood, metabolize nutrients and help pass out waste products from filtered blood.

In the nephron, a complicated chemical exchange takes place between capillaries and urine-carrying tubes. As a result waste materials and water leave blood and enter urinary system.



The excretory product removed from each kidney passes through the ureter to the urinary bladder. The urinary bladder functions to store this excretory product temporarily. When it is full, the excretory product will be eliminated out of the body through the urethra. This excretory product is discharged in the form of yellow-coloured liquid known as urine.

### KIDNEY MALFUNCTION

- ✓ Investigate the possible causes of Kidney malfunctioning.
- ✓ Suggest techniques to cure problems of kidneys.

Kidney failure happens when they do not work properly. This could be due to some underlying diseases such as diabetes, hypertension

or infections caused by microorganisms. A blockage in ureter and urethra in the form of calcification or stone can cause damage to the kidneys because urine cannot flow out of the body. A person who has only one kidney still can live normally. If both kidneys fail, the person will have to rely on an artificial kidney machine called the dialysis machine to remove wastes from the blood.

## Dialysis

- The process of filtering and cleansing the blood outside the body is called Dialysis. It is done by using a dialysis machine which purifies blood by two distinct processes 1) filtration and 2) reabsorption.
- Dialysis removes excess water and wastes through a semi-permeable membrane (which allows certain substances to pass through it but not others) and works on the principle of osmosis and diffusion.
- A semi-permeable membrane has pores which allow urea and salts to pass through.
- Larger particles such as blood cells, bacteria, viruses or protein cannot pass through this membrane.
- The common dialysis method is hemodialysis where waste products and excess water are separated artificially from the patient's blood.

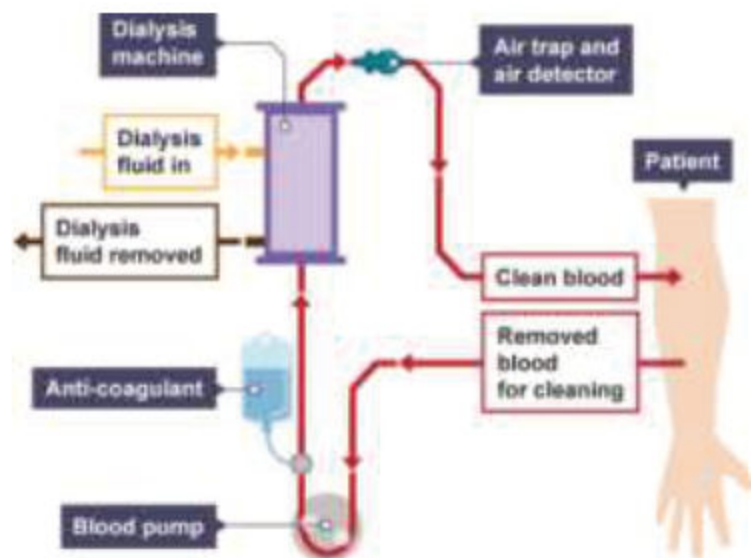


Figure. 1.11: Process of Dialysis

## Kidney Transplant

The acute renal failure can be managed by kidney transplant. In a kidney transplant a healthy kidney received from a donor is placed inside the body to do the work which cannot be done by patient's own kidney.

### DO YOU KNOW?

**Role of Sindh Institute of Urology and Transplant (SIUT):** Sindh Institute of Urology and Transplantation (SIUT), located in Karachi, Pakistan is one of the most reputed medical institutions in the South Asian region, SIUT is providing free of cost specialized medical care and transplant procedures to kidney and liver patients. This all have been possible with selfless leadership and spirit of service for humanity by Dr. Adeeb ul Hassan Rizvi.

## SUMMARY

- Main control system of our body is Nervous System.
- It consists of three main parts 1. Brain 2. Spinal Cord 3. Nerves
- Brain, an organ of soft nerve tissue, is located in upper part of bony head called skull.
- Brain is divided into three parts 1. Cerebrum or fore brain 2. Cerebellum or mid brain 3. Hind brain.
- Pons and Medulla Oblongata with the mid brain are often called the brainstem.
- The human nervous system serves three main functions: 1) collect information from sense organs; 2) integrate information by processing and evaluation 3) decide to take action.
- Water, CO<sub>2</sub>, salts and metabolic waste like Urea are body wastes which are expelled from the body.
- The main organ of the urinary system includes the Kidney and its functional unit, the nephron.
- The water and urea are expelled out in the form of urine through the Ureter.
- The kidney can be damaged due to microbial infection, calcification and other diseases.
- The kidney disease or failure can be managed through dialysis and kidney transplant.

## EXERCISE

### 1. Fill in the blanks

- i) Upper part of the bony head is called \_\_\_\_\_.
- ii) Top part of the brain is known as \_\_\_\_\_.
- iii) Cerebellum means \_\_\_\_\_.
- iv) Hind brain consists of pons and \_\_\_\_\_.
- v) The largest part of the brain is \_\_\_\_\_.
- vi) Extension of the brain through vertebral column is called \_\_\_\_\_.
- vii) The waste material from the process of cellular respiration is \_\_\_\_\_.
- viii) The main excretory organ is \_\_\_\_\_.
- ix) The artificial kidney machine is called \_\_\_\_\_.
- x) Acute kidney failure can be managed by \_\_\_\_\_.

### 2. Circle the best answer

- i) The nervous system is network of:  
a) Neurons b) spinal cord c) nerves d) lobes
- ii) Fore brain is called:  
a) Cerebellum b) Cerebrum c) Medulla Oblongata d) Pons
- iii) The central nervous system is composed of:  
a) Cerebrum, Thalamus and Hypothalamus  
b) Cerebrum, Pons and Medulla oblongata  
c) Brain, Spinal cord and nerves  
d) Cerebellum, Brain and nerves
- iv) The main excretory organ is:  
a) Kidney                      b) liver                      c) skin                      d) lungs
- v) Replacement of malfunctioning organ by a healthy organ is called:  
a) Transformation                      b) Transplantation  
c) Transpiration                      d) Translocation

### 3. Answer the following question

- I. What does nervous system do? How does it transmit messages between the brain and the body?
- II. List some of the things your nervous system controls in your body.
- III. Nervous system has three main parts. What does each part do?
- IV. Name the parts of the brain which control movement, thinking, emotion, breathing, growing and temperature?
- V. Describe the functions of:
  1. The frontal Lobe                      2. The parietal Lobe
  3. The Temporal Lobe                4. Occipital Lobe
- VI. Differentiate between voluntary and involuntary movements.
- VII. Give five examples of reflex action related with daily life.
- VIII. Why does the body need an excretory system?
- IX. Explain the role of kidney in the excretory system.
- X. Draw and label Human Excretory system.

### PROJECT WORK

- Compare brain with computer, Discuss with your classmates.
- Visit your nearby hospital and interview a neurologist to find out the common brain disorders in the area and discuss some ways to cure them. Also, find the rate of death due to brain injuries in road accidents.
- During your visit, interview some kidney patients to find out the process of dialysis. Also, ask from doctors as how many kidney patients go to kidney transplant per year.

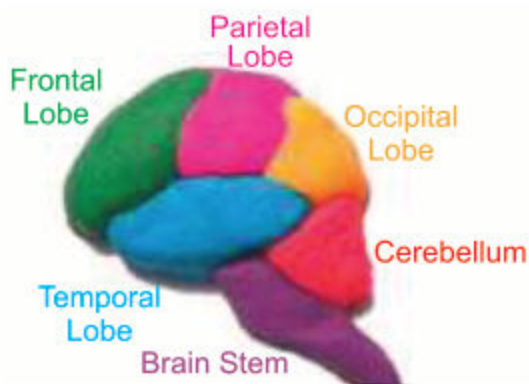
### THINK QUEST

1. Why Brain is the boss of human body?
2. Why our brain works all the time and does not stop working during sleeping?
3. What is the function of spinal cord in Nervous System?
4. Why do doctors usually suggest liver function test when diagnose the kidney related problems?
5. People often sweat more while exercising. One reason is because sweating helps us to stay cool. Give another reason for that.
6. There are people, who have been born with one kidney, yet they are normal. What precaution they should still take?

## MODEL MAKING

### Activity: Making a model of Brain (Group work)

**Material Needed:** Modeling dough/ Plaster shine/ clay in 5 different colours/ (mix different food colours in it)



Teacher note: Teacher should observe students and make sure that everyone participates actively. After the completion of model, teacher should ask students to remain in the same group and give whole class presentation on function and structure of nervous system with the help of the model they have prepared. Ensure active participation of each group member by dividing the role and responsibilities among group members.

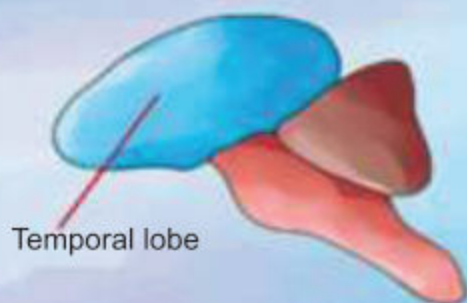
| PROCEDURE                                                                                                                                                                                                                                                                                                           | DIAGRAMS |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><b>Choose 6 different colours of playdough to create a model of the brain as shown in the picture:</b> Each color will be used to create different lobes of the brain. Using different colors will make it easier for you to separate and identify each part of the brain. (Use colours of your own choice).</p> |          |
| <p><b>The brain stem:</b> Take a small amount of playdough / clay and roll it between your palms to create a thick rope. It should have a slight "s" or elbow shape as shown in the figure. Smooth the rope with your fingers until the top of it curves upward and to the</p>                                      |          |

left, while the bottom should be slightly longer than the top section and sink off to the right. The bottom should also have a pointed tip, while the top should have a flat edge and look a bit wider overall.

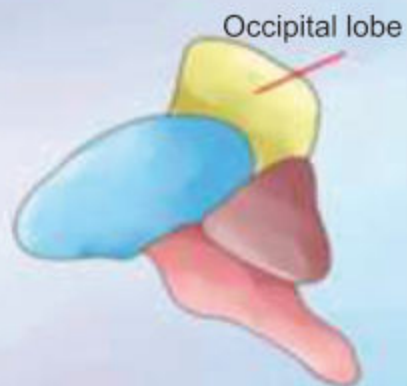
**Attach the cerebellum:** Take pinch off roughly half the amount you used with creating the brain stem. Roll and form this into a triangle with rounded edges. Position it so that one side of the triangle sits in the upper curve of the brain stem.



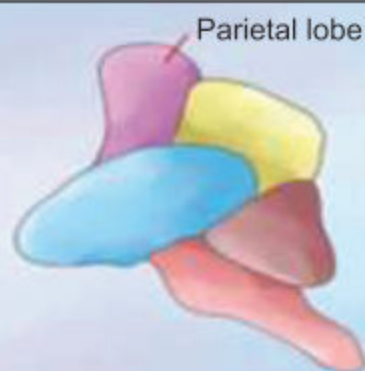
**Create the temporal lobe:** Pinch off a roughly the same size lump of clay you used for the brain stem. Roll this clay into an oval shape. Place the center of this oval onto the top of the brain stem and press it gently to attach the two pieces of clay together. The bottom left half of the oval should reach half way up the left side of the cerebellum if we consider the bottom of the triangle to be the part matched to the brain stem.



**Move on to the occipital lobe:** Take a piece of clay roughly the same size as your temporal lobe. Roll and flatten this piece into a quadrilateral which inclines out at the top making a muffin shape. Position it so that the bottom center of the lobe connects to the top left 1/4 of the temporal lobe. The right side of the occipital lobe should cover the other left half of the cerebellum, with the muffin top spilling slightly over the top of the cerebellum.



**Add the parietal lobe:** Pinch off slightly more clay than the amount used to create your occipital lobe. Form another rectangle just larger than a square. One bottom shorter edge of the rectangle should cover the rest of the right half of the oval made by the temporal lobe. The rectangle should lean slightly to the right.



**Make the frontal lobe to complete the atlas:** This should be your largest bit of clay, and slightly larger than your original amount used to create the brain stem. Roll it into an oval, then slightly flatten the bottom right and left sides to attach them to the rest of your brain atlas. Wedge this final piece in place on the left of your model to create the front of the brain. The flattened bottom right section will attach to the parietal lobe, while the bottom left covers the top left half of the temporal lobe oval slightly overlapping the edge.

